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University of Colombo, Sri Lanka

UCSC

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination — Semester II - UCSC AY19 [held in March/ April/ May 2023]

SCS 1211 – Mathematical Methods - I

(Two (2) Hours)

Answer ALL questions

Number of Pages = 4

Number of Questions = 4

Important Instructions to candidates:

I. Students should answer in the medium of **English language only** using the answer paper provided.

II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.

III. Write your index number **CLEARLY** on each and every page of the answer paper.

IV. This paper consists of **4** questions in **04** pages (including the Cover Page).

V. Answer **ALL** questions.

VI. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are **not allowed**.

VII. **Non-Programmable** calculators are **allowed.**

1.

a. Let $A = \begin{bmatrix} 1 & 2 & -2 & 2 & 3 \\ 1 & 2 & -1 & 3 & 5 \\ 1 & 2 & -3 & 1 & 1 \end{bmatrix}$.

i. Using elementary row operations, reduce A to **row reduced echelon form.**

[8 Marks]

ii. Find the rank of A.

[2 Marks]

iii. Solve the following system of linear equations:

$$\begin{aligned} x_1 + 2x_2 - 2x_3 + 2x_4 &= 3 \\ x_1 + 2x_2 - x_3 + 3x_4 &= 5 \\ x_1 + 2x_2 - 3x_3 + x_4 &= 1. \end{aligned}$$

[7 Marks]

b. Let A $= \begin{bmatrix} 1 & 2 & 2 \\ 1 & 0 & 2 \\ 3 & 1 & -1 \end{bmatrix}$. Find the A^{-1} using Gauss-Jordan Method.

[8 Marks]

2.

a. Let $A = \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}$, where $a, b, c \in \mathbb{R}$. Using the properties of determinants,

show that $\det(A) = (a-b)(b-c)(c-a)$.

[7 Marks]

b. Find the redundant vectors in the following sequence of vectors in $\mathbb{R}^3$. Write each of the redundant vectors as a linear combination of non-redundant vectors:

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix},\ u_2 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, u_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \textit{and } u_4 = \begin{bmatrix} 1 \\ 6 \\ 1 \end{bmatrix}.$$

[7 Marks]

c. Let $P_3(\mathbb{R})$ be the vector space of polynomials of degree at most 3. Determine whether the following sequence of vectors forms a basis for this vector space:
$x + 1,\ x^3 + x^2 + 2x,\ x^3 + x^2 + x,\ x^2 + x.$

[6 Marks]

d. Let U, W be subspaces of a vector space V. Let $U + W = \{u + w\ [u \in U, w \in W\}$. Show that $U + W$ is a subspace of V.

[5 Marks]

3.

a. Let $T: V \longrightarrow W$ be a linear Transformation, where V and W are two vector spaces over the same field $\mathcal{F}$.

i. What is meant by the null space of T? Shaw that the null space of T is a subspace of the vector space V.

[4 Marks]

ii. What is meant by the range of T? Shaw that the range of T is a subspace of the vector space W.

[4 Marks]

b. Let $T: \mathbb{R}^3 \longrightarrow \mathbb{R}^4$ be the linear transformation defined by

$$T\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 - x_2 + x_3 \\ x_2 - x_3 \\ x_1 \\ 2x_1 - 5x_2 + 5x_3 \end{bmatrix} \quad \text{for all } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3.$$

i. Find the range of T ($\mathcal{R}(T)$) and a basis for $\mathcal{R}(T)$.

[5 Marks]

ii. Find the null space of T ($N(T)$) and a basis for $N(T)$.

[5 Marks]

c. Let $T: V \longrightarrow W$ be a linear transformation, where V and W are two vector spaces over the same field $\mathcal{F}$. If V is a finite dimensional vector space, prove that $\dim(\mathcal{R}(T)) + \dim(N(T)) = \dim(V)$, where $\mathcal{R}(T)$ and $N(T)$ are the range and the null space of T.

[7 Marks]

4.

a. Let $O = \{u_1, u_2, \cdots, u_n\}$ be a set of n vectors in a real inner product space V. When do we say that O is orthogonal?
If O is orthogonal, show that O is linearly independent.

[6 Marks]

b. Let $v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 3 \\ 1 \\ -1 \end{bmatrix}, v_3 = \begin{bmatrix} 2 \\ 4 \\ 2 \\ 2 \end{bmatrix}$ be three vectors in $\mathbb{R}^4$. Find three orthogonal vectors u_1, u_2, u_3 such that $\langle \{v_1, v_2, v_3\} \rangle = \langle \{u_1, u_2, u_3\} \rangle$.

[9 Marks]

c. The following table shows the average day temperature and number of young coconuts sold in a city for seven days:

Average day temperature	**Number of young coconuts sold**
17°C	320
22°C	570
26°C	850
20°C	470
24°C	750
23°C	620
23°C	680

Find a function of the form $y = Mt + C$ that best fits the data, where t is temperature in degrees Celsius and y is the number of young coconuts sold.

[10 Marks]